

CHAPTER 9

PASSING

PASSING MAY SEEM LIKE nothing more than just putting your foot down and steering around the car ahead of you, but it's not

that simple. The uninformed observer misses the point that on the racetrack, *everyone* has their foot on the floor. You even hear shades of this misconception from television race commentators. One driver will be methodically catching another and the announcer says something like, "Wow, Bobby, Stroker Ace is really putting the hammer down now," as if he were driving around on half throttle for three quarters of the race and only just now decided to go full out.

The pass of choice for anybody that ever put on a helmet is the "Big Horsepower Blow-By." As Walt Bohren, Skip Barber instructor and two-time IMSA GTU Champion, says, "You *can't* have too much horsepower." The people who make the rules, however, spend a lot of time designing regulations that make it difficult for any one competitor to get this kind of advantage. By restricting the weight, engine size, and shape of racecars, the sanctioning bodies try to limit the possibility that any one team can get a huge technical advantage over the others.

The result is that in any well-contested class of racing, racers in the top 25% of the field have little or no performance advantage over each other. Passing then becomes a matter of being able to wring more performance out of the car than your competitor: a contest of driving skill.

THE RULES

Although the racetrack can seem like a wild place where chaos reigns and the competition between strong egos knows no bounds, there are some conventions that are observed to prevent total anarchy. For example, it is the overtaking driver who has the primary responsibility to do the overtaking without making contact with the car being passed. The major sanctioning bodies for road racing in the U.S. agree that the overtaken driver also has the responsibility

Passing on the racetrack requires an understanding of the conventions in play and the techniques required to improve your position or to fairly defend it.

to help make the pass a contact-free affair.

The primary problem occurs not on the straightaway, where driving on any one

part of the straight holds no advantage over another part, but in corners, where everyone wants to be on-line. The most important question in passing situations is who's got rights to the line and who will yield.

Rights to the Line

Our standard for rights to the line hinges on the relative positions of the cars at the approach to the corner. The SCCA rulebook says, "The responsibility for the decision to pass another car rests with the overtaking driver." If, at the brake point, your car is dead alongside the car you're trying to pass, the overtaken car is obliged to leave room for you and not simply turn into you at the turn-in point. If, however, you're trying to pass but you're not quite alongside, as in having the front wheels up to the middle of the other car at the brake point, you haven't made the pass; it's your responsibility to get out of the way of the other car and yield the line in order to avoid a collision. The other driver might yield and give you room at the apex, but you can't count on it. He may not even have seen you



Fig. 9-1. If the car attempting the pass is alongside at the approach to the corner, the car being overtaken is obliged to yield.



Fig. 9-2. If the overtaking car is only partway alongside on the approach to the corner, the convention is that the car in the lead has the right to the line and the burden is on the overtaking car to avoid contact.

if he hasn't been looking in his mirrors. If you're not alongside, you've got to expect to be "chopped," which is the racing term for another driver pulling across the nose of your car going to the apex.

"I go with the conventional way of looking at rights to a corner. I agree 100% that the guy getting into the corner first has the corner. Some people don't see it that way. If you're trying to pass me, and at the point of turn-in I've got you by half a car length, then I've got the lead going into the corner and I'm going to go for the apex."

—Dorsey Schroeder

Pushing the Envelope

If you watch races on television, you can see plenty of examples of passes attempted by top-level professionals that lead to crashes rather than changes of position. Invariably, they are caused by the driver behind forcing the issue, trying to complete a pass by putting the burden of accident avoidance on the overtaken driver. If the overtaken driver isn't in the mood to be bullied out of his position, the two cars crash. We don't have to name names, but if we did, it would include some of the top names in the sport.

Before the cars and racetracks became as safe as they are today, cars making contact, especially open

wheel cars, often led to fatalities. The drivers raced hard but they passed because they drove faster, not because they were willing to run into other cars. Perhaps the relative safety of the cars and circuits today has made drivers less protective of each other's lives, since most of the time a crash only tears up equipment rather than killing a fellow competitor.

Different Rules for Different Arenas

We put on over 200 individual Formula Dodge races a year, and in these races, we have made a clear distinction between a driver who causes an accident in a passing situation (and pays for the damage to both cars) and the victim. If the overtaking driver is not at least alongside at the turn-in point (at the latest) and there is contact between the cars as a result of the pass attempt, the overtaking driver is at fault. If the overtaking driver is alongside and the overtaken driver turns into him, the blame is on the overtaken driver.

Our rules work pretty well with cars that have comparable corner entry speeds, but the situation is different if one car approaches and enters the corner 20 m.p.h. slower than another. Sometimes, especially in a long corner, the faster car can be significantly behind the slower one at the braking point, yet because of the high closing rate, get safely by before the apex. You'll run into this situation at races in which multiple classes compete. Because of the varying speed potential within the same race, the pro rulebook puts responsibility on both drivers to assure safe passing.

The rulebook reads, "It is the responsibility of both the overtaking and the overtaken drivers to assure safe passing at racing speeds....However, if it [the overtaken car] is overtaken by a faster car, the driver must give way to the overtaking car."

It can be confusing for novice drivers to sort out their rights and responsibilities in passing. In many forms of racing, a good pass is one you get away with. Still, there are guidelines and conventions that help sort out the conflicts. We think you can break down most situations into the following three categories:

I. Equal Cars Racing for Position.

1) If you are attempting a pass and are alongside at the braking point, you have rights to the line from the turn-in to the apex. The overtaken car should yield.

2) If you're not quite alongside at the brake point but draw alongside before the turn-in point, you technically have rights to the line, but if contact happens, you're really to blame. You are taking the chance in this situation that your guesstimate of the closing speed is correct—that you'll make up the distance of a quarter car or half car between the brake point and the turn-in. If you guessed wrong and the other car is still half a car-length ahead at the turn-in, expect that the other driver will take the line and go for the apex. You didn't do the pass—you got in the way.

3) If you're behind the other car and attempt to make up a car-length or more between the brake point and the turn-in, you're either a wild optimist or a lot better braker than the other driver. Either way, if you blow it, it's your fault.

II. Equal Cars where One is Being Lapped or is Uncompetitive.

1) If you're driving the slower car and the faster one is alongside you in the braking zone, you should yield the line.

2) If the faster car is not quite alongside you, you should still yield the line.

3) If the faster car is behind you at the turn-in point, you should, as a courtesy, leave one and a half car-widths between your car and the apex, letting the other driver decide whether to do the pass in the corner or on the following straight.

III. Races with Multiple Classes of Car.

The big difference in this type of race is that you may be racing for the lead in your class when another car, or group of cars, appears in your mirror, about to lap you. Typically, they are racing for the lead in *their* class, going 5 seconds per lap faster than you. You don't want to hold them up by getting in the way, but you sure don't want to slow up and lose time to the competitors in your own class.

In professional sportscar racing, where this happens frequently, the rulebook states that you should give way to the faster class of car. This doesn't necessarily mean that you pull off the racetrack, you just

make it easier for the faster cars to get by without overly handicapping yourself. The faster cars should have no trouble getting by on the straights even if you stay on the normal line. Ideally, they would like to pass you and get back on-line before the next corner, and when the timing works out this way, passing is no problem. Just as often, the faster cars don't get by you on the straight but are closing ferociously in the braking zone and corner. If you see this developing and you're racing hard for position in your class, give them the car-width and a half at the apex. You won't have to slow down too much to make this happen without spoiling the race in either class.

If you are uncompetitive in your class or running a minute behind the car ahead and a minute ahead of the car behind, you can spare more road—and more lap time—to help faster cars through.

On some circuits, the race organizers may ask the drivers of slower classes of cars to keep to a certain lane on the high-speed straights or on the banking. You'd be expected to finish the corner leading onto the straight by tracking out to the edge, then working your way carefully to your lane. At the end of the straight you can work your way back out to the turn-in point if there is no traffic. If a faster car is going to get there at the same time, you'll have to stay where you are and enter the corner off the optimum line, which is going to require an adjustment to your braking and turn-in points.

When you are yielding to faster cars, unless specifically instructed by the race organizers, drive the normal line down the straights. Be sure to check your mirrors at the beginning and end of every straight. If you get caught at the corner, give the faster cars a car-width and a half from the turn-in through the apex to allow them to choose to go by in the corner. If, on checking your mirrors, you judge that they won't catch you until the following straight, stay on-line through the corner so as not to lose any time to your competitors.

COOPERATING

In all of these iffy passing situations, it takes a lot of the anxiety away from the overtaking driver if it's obvious that the slower car ahead is going to yield the line. Wondering whether the driver alongside is going to yield or turn into you is uncomfortable.

The Point-By

One way to alert the faster driver that you're yielding is to point to the side where you're going to leave room. This move is known as the "point-by," and it isn't a command for the overtaking driver—it's merely a suggestion. The overtaking driver may have already made the decision to pass on the other side, so don't be surprised if you point one way and are passed on the opposite side. For the passing driver, the value of



Fig. 9-3. The point-by takes the worry out of passing situations and is an indicator of a heads-up driver displaying sportsmanship.

the point-by is not so much the suggested direction but the fact that you know he or she is around.

Don't point if you don't mean it. If you're at the turn-in point and you point inside to signify that you're prepared to leave room at the apex, don't then go for the apex and shut the door. The overtaking driver may not have gone for the pass in the corner except that you encouraged it by pointing. Going for the normal line has the effect of suckering in, then taking out your competitor. We've seen it many times: in fact, we've named it the "Point-Bomb."

"In my type of racing it's amazing that the guys generally do a really good job of point-bys. You try to help that by showing your appreciation. Every car I ever passed I've stuck a hand out the window and waved back if I could. It establishes a communication between drivers, and a respect—and that works real well, especially in endurance racing where you're going to see each other again and again."

—Terry Earwood

Driving Off-Line

You can also help faster drivers get around by leaving the normal racing line so that they can get by on-line. However, slower drivers, especially novices, can overdo it in an effort to be cooperative. As a faster driver approaches a slower car he or she is watching to see where the slower car drives. The faster driver plans a pass for a part of the racetrack where it's possible to get by while losing as little lap time as possible. The faster driver is planning on your car being in a particular spot, and if you drastically alter your line

in an effort to cooperate, you may end up right in the spot the faster car had committed to go, causing more problems than you solve.

The best procedure for a slower driver is to drive the same line lap after lap, allowing the drivers behind to plan their passing moves. If you do make a departure from the line, make it a subtle one: one and a half car-widths off the apex, for example. Don't try to help by braking 100 yards sooner than you did in the last lap. The driver behind would never expect that drastic a change in braking point, and you'd surely get punted from behind. It pays to be consistent.

PASSING EQUAL CARS

Two cars that have the same speed potential make passing difficult. The easy pass—one car going a lot faster than the other on the straight— isn't going to happen. With the same acceleration potential, both cars' top speed should end up the same, provided they start at the same speed. If one driver does a better job than the other through the corner leading onto the straight and starts the straight at a higher speed, that car will gain ground on the other.

The Exit-Speed Pass

One of the basic laws of passing is that in order to pass another car, you need to be going faster than it. That concept doesn't sound too complicated, yet frequently we see drivers pull out from behind another car going exactly the same speed, expecting to pass it. The only thing accomplished is two cars driving down the straight in formation, doing a fly-by of start/finish.

In order to come onto the straight faster than the driver ahead, you can't put yourself in a position where the car ahead can determine your cornering speed. The most common mistake we see in passing technique is a driver that follows the car ahead so closely that he or she can't go any faster without hitting it. The driver then bitterly complains that the driver ahead was blocking, slowing up the run



Fig. 9-4. Setting up an exit-speed pass requires leaving room ahead to accelerate as the cars come to the corner exit.



Fig. 9-5. Staying right on the gearbox of the car ahead allows that driver to dictate cornering speed.

through the corner. The fault, of course, is on the driver that allowed the block by not leaving enough room so that he or she could take the corner faster.

In order to pass using an exit-speed advantage, you need room to accelerate, not allowing the slower car ahead to impede your progress. You would like to time it so that you don't catch the car ahead until you're on the straight. At that point, for example, a 2-m.p.h. advantage gained from doing the corner better would convert to 3 feet per second. Exiting the corner at that faster rate, you'd be alongside a 12-foot long car in 4 seconds, and a car-length ahead in 8 seconds.

How Drafting Increases Speed

A lot of your engine's horsepower is used to push the air out of the way of the car: in a Formula Dodge, for example, over 80% of the car's total horsepower is being used to overcome the aerodynamic drag of the car at speeds in the 100-m.p.h. range. Most of the rest of the engine's horsepower is used up by overcoming transmission frictional losses, the rotational inertia of all the moving components, and the tires' rolling resistance. Eighty percent of the resistance to going faster at the end of the straight is aerodynamic and, therefore, if this drag is reduced, more of the horsepower is available to accelerate the car; as a result, it goes faster.

Two cars in line, regardless of their shape, present a more streamlined shape to the air and as a result, both have lower form drag. The longer the line of cars, the greater the form-drag reduction and the faster the whole train will go.

With equal racecars, the car ahead is at a disadvantage—even though it is getting some benefit from the draft—because it is the one punching the hole in the air, while the one following behind gets a speed advantage. The draft-by pass involves the driver behind using this speed advantage to close on and pass the car ahead.

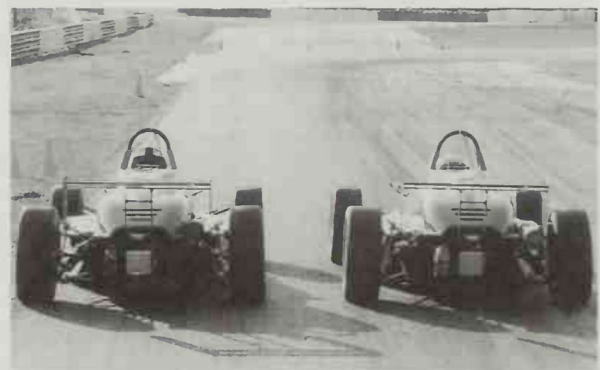


Fig. 9-6. The properly executed draft-by pass starts by leaving room ahead to accelerate into (top), then smoothly transitioning out from behind the car ahead (middle) and using the speed advantage to pull alongside in the braking zone (bottom).

Phases of a Draft-By Pass

In using the draft to pass a single car, you should think about each aspect of the pass. The first step is to leave some room ahead. You'll probably want to draft by on the fastest parts of the course where the draft is great-

er, so you'll pick a corner leading onto one of the high-speed straights as a place to set up the pass.

If you're right on the gearbox of the car ahead, you'll need to leave some room—one, two, or three car-lengths—so that you can use the draft to close on the car ahead coming onto the straight. You'll try to time it so that you come off the corner faster and get a run down the first third or half of the straight, never having to lift off the throttle, building the speed differential that will carry you by. As you can see in Fig. 9-7, the relative vacuum behind the car ahead allows you to go faster.

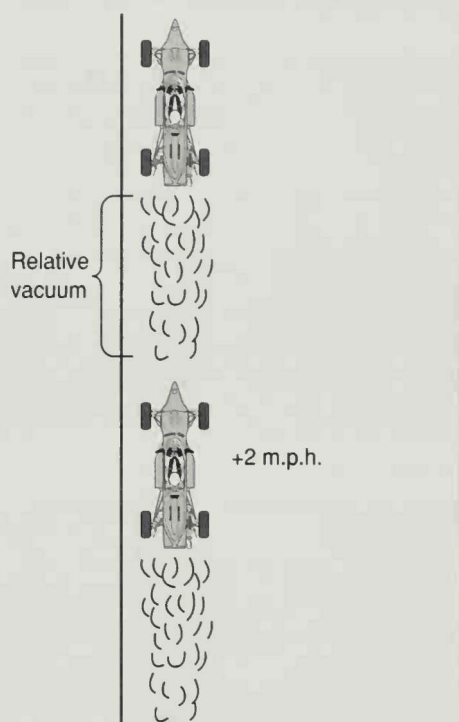


Fig. 9-7. The following car gains speed because its aerodynamic drag is reduced in the relative vacuum behind the lead car.

As you get closer (see Fig. 9-8), your car smooths out the turbulent wake behind the lead car and it also sees a reduction in drag and gains speed, although, since you are closing on the car, your advantage is greater.

When you're one to one-half car-lengths back, you steer to the inside to drive around. When you leave the spot directly behind the car ahead you no longer smooth out the airflow for the leading car. As you can see in Fig. 9-9, the turbulence reappears behind the lead car, making the drag higher and slowing its acceleration. This can be a big problem for both cars if the passing driver doesn't anticipate the change in closing speed.

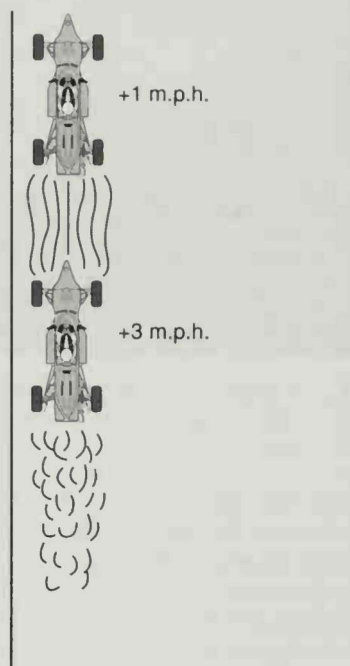


Fig. 9-8. As the following car gets closer, it smooths out the turbulence behind the lead car. Both gain speed, although the following car maintains its previous speed advantage.

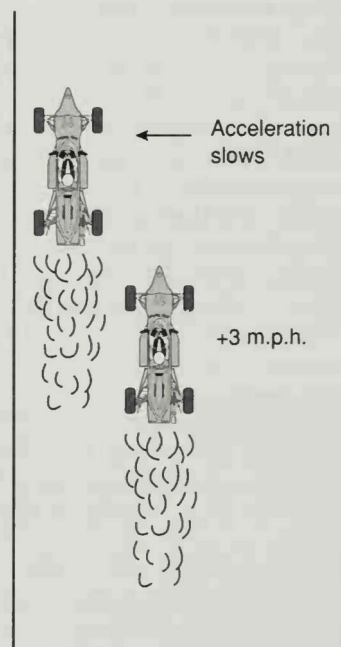


Fig. 9-9. When the following car pulls out from behind, the turbulence reappears behind the lead car, slowing its acceleration.

"At the Glen, running by myself, I'd see 125 to 126 m.p.h. at the six hundred marker at the end of the straight. When Scott Gaylord came out in another of our cars, I tagged on to him and I saw 131 on the speedometer. 5 m.p.h.! Free horsepower! The tow was so good that if you didn't pull out to pass, you had to run two-thirds throttle all the way down just to keep from hitting him from behind."

—Terry Earwood

Watch the Changing Closing Rate. As you approach the rear end of the car ahead, getting a run on it, you watch the closing rate between the two cars and time the pullout so that you miss the lead car's rear corner with your front corner as you go by. Because the drag reappears for the front car when you pull out from behind, the closing rate between you accelerates—the car ahead seems to dart back at you. To avoid tapping the car ahead as you pull out of the draft, a smooth pullout of the relative vacuum behind the car ahead works best. Remember too that you want to turn the steering wheel as little as you can get away with. Big jerks of the wheel will only scrub speed away—the more you turn the wheel, the more drag the tire creates against the road.

Another reason to avoid jerking the car out of the draft is that the hole in the air made by the front car doesn't only extend directly behind the car. A car pushes the air aside much like a boat pushes water aside, creating a bow wave and a wake. There is still draft advantage to be had alongside the rear quarter of the car (see Fig. 9-10).

Close, Pull Out, Pull Back. Once you've pulled out of the wake of the car ahead, it would be ideal if you had gained enough speed that momentum would carry you completely around. This isn't always the case since, once you're back alongside the other car, all the drag that you would have had running alone reappears. Your car doesn't instantly slow to its original one-car top speed: the speed advantage you gained from the draft deteriorates as you pull alongside. If you had your choice, it wouldn't fall to zero before you got completely past and pulled back in line, but it doesn't always happen this way.

Sometimes your car runs out of steam when you're 7/8ths past. You can't pull back in line until you've checked in your mirror that the car you just drafted is indeed behind you. Just because you saw his nose at your shoulder three seconds ago does not mean that you're completely past. Your speeds may have equalized since then and you still need a few feet to get completely ahead. Move over without looking, and you risk contact—on the fastest part of the circuit.

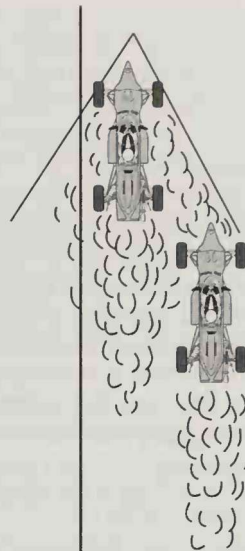


Fig. 9-10. A relative vacuum also exists alongside the rear quarter of the lead car.

If you're in the car that just got drafted, you have to realize that the driver of the other car is susceptible to this kind of mistake and be prepared for the worst. You might even help him along by breathing out of the throttle and letting him tuck back in front. Remember that both of you will have less drag in-line than running side by side. If there are others in the race, as is most often the case, the two of you can go faster than any other single car if you work together.

"A successful draft-by pass starts before the corner leading onto the straight. You have to slow down the entry and time it so you do the corner as fast as you can, but never get held up by the driver ahead—you know, leave one, two, or three car-lengths ahead of you. It's hard, especially for new drivers to have the confidence to drop back a bit. Getting the run is what makes it happen."

—David Loring

In this case you've been drafted and almost passed by the other guy—there's no possible advantage in sticking to his rear quarter. When you get to the corner, he'll have to drive off-line because you're in the way at the turn-in. You'll have to drive off-line because he's in the way at the apex. By surrendering three feet of room and letting the car back in front,

you lower each other's drag *and* you're able to run through the next corner on-line—the fastest way. By being difficult, you slow both cars up and give an advantage to the drivers behind who are chasing you.

Planning on the Draft-By

There are too many variables involved in passing to be able to predict whether you'll be able to do a clean, unimpeded draft-by pass on the straight. For certain, it's one of the easiest passes—almost as good as the Big Horsepower Blow-By. It doesn't compromise your line through the corners and, if anything, should make you faster down the straights. Whether you can do it depends upon a lot of things. For example, the racetrack has an effect on your passing technique: if the track has short, slow straights, it isn't going to be good for drafting and it won't play a big part in passing strategy.

Horsepower also plays a part in strategy. A high-horsepower class of cars, where the acceleration is still strong at the end of the straight and the aerodynamic drag absorbs a smaller percentage of the overall horsepower, won't show as dramatic a draft effect as lower-horsepower cars.

In terms of car design, some racecars are extremely slippery, making a smaller, less disturbed hole in the air and consequently reducing the advantage to cars behind. You can try to draft off these cars, but nothing will happen, since the reduction in drag is so slight that your car can't gain a usable amount of speed.

But sometimes the car and racetrack combine to create a killer tow, and driving by your competitor on the straight every other lap is a cinch, as long as you can stay within six or seven car lengths on other parts of the course. Of course, if you can do it to others, they can do it to you; often, you'll see cars swap positions on successive laps all race long, each using the draft advantage to get by the other. If neither driver can pull away on other parts of the circuit, the race comes down to the position of the cars at start/finish on the last lap and strategy plays a big part in who wins.

PASSING IN THE BRAKING ZONE

If neither the draft nor better exit speed provides you with enough speed advantage to actually pass the car ahead on the straightaway, you'll have to complete the pass under braking. Being able to threshold brake at the last possible moment when you want to or need to is possibly one of the most important skills for racing. It's the way you end up doing a lot of your passing, and it's another vital weapon in your arsenal.

Don't Pull Over

The idea of a corner entry pass is to change the relative position of the cars. At the turn-in point of the corner, you are going to be off the normal line by at least a car width. In order to avoid early apexing the

corner, you'll have to make an adjustment to your turn-in point.

Since you're going to be forced to drive off-line, the lap time for the lap on which you do the pass is going to be slower than normal. Giving up a car width of radius in the first part of the corner is going to limit your cornering speed in comparison to the normal, bigger radius line. This is okay, since here, you're interested in changing position, not lowering lap time. At the same time you want to minimize the sacrifice you make, surrendering as little lap time as possible. To do so you want to get back to the normal racing line and up to your normal corner exit speed as quickly as possible.

Your goal is to be back on-line and on power by, at the latest, the apex. If you don't manage this, you'll lead going into the corner but get passed by your competitor going out.

"I brake as late and hard as I feel comfortable doing, all the time, but there are times when I end up going beyond that. That's the difference—you can extend yourself beyond what your comfort level is, but for a reason."

—Bryan Herta

The Perfect Corner-Entry Pass

Let's follow the progression of a corner-entry pass that works. At the normal brake point the cars are side by side. Both drivers lift off the throttle and go to the brakes. The driver on the inside has gotten alongside and established the right to the line at the apex according to the generally accepted rules of the road.

The Driver on the Outside. In most cases it is in the best interests of the driver on the outside to yield to the overtaking driver for the following reasons. First, as you learned earlier, the car on the inside has the advantage of the laws of physics. That is, the outside car has a longer way to drive around the outside and, given cars with equal cornering power, the car on the outside will lose ground to the inside car as they drive through the corner. Second, the driver outside is off the normal racing line and the asphalt is likely to be more slippery out there, making the car corner worse than the car on the inside. Third, if the inside car should slide wider than intended at the exit of the corner, the outside car will be the first to end up on the dirt—or whatever lurks outside the track-out point.

The decision to yield does have something to do with where in the race the pass is taking place. If it's the fourth lap of a four-hour race and the driver on the inside is one with whom you expect a long seesaw race, yielding is no big deal, since you'll probably get ahead on the next lap or the lap after that.

If it's the last lap and the finish line is 50 feet past the exit of the corner—well, that's something else. In a mid-race situation you concede the corner if the other driver is alongside in the braking zone. If anything, you brake a little bit early and plan to get in line behind the other car. As you'll soon see, there are mistakes the overtaking driver can make that may put you back in front in short order. In order to be prepared to take advantage of any mistakes on the other driver's part, you want to nail the turn-in point and be perfectly on line. Next, you want to be set to attempt to get the throttle open a little earlier than normal if the opportunity presents itself. If the overtaking driver does the pass well, you can expect that you won't be able to apply throttle until the car ahead does, but if the other driver makes a mistake and is not directly in front of you as you approach the apex, you may be

"Going in deeper on the brakes in a passing situation is probably the most difficult thing to judge in a race car. You get a draft on a guy and you are coming into the braking zone with more speed—where do you put on the brakes? You want to go deeper than the other guy but you're also on a piece of road you're not normally on. It's the toughest thing to judge."

—Jeremy Dale

able to get an exit speed advantage by beating the other car to the acceleration away from the corner.

The Overtaking Driver. The overtaking driver, in order to give away as little radius at the corner entry as possible, does not want to give the car being passed any more room than necessary. A common mistake is for drivers to be very gun shy of the car next to them and to leave two or three car widths of room between them in the braking zone. All you need is two feet in open wheel cars, less in cars with fenders.

You also want to brake parallel to the edge of the racetrack. Many drivers, feeling edgy about the car next to them, head diagonally toward the apex, opening up the possibility of early apexing, which is going to cause trouble at the corner exit.

Sketching out the approach to the corner, the two cars are side by side (see Fig. 9-11). The car on the inside should intersect the normal racing line by the throttle application point or, at the latest, the apex. To do this, the turn-in point moves down closer to the corner to point A. The distance between the normal turn-in point and the new, later turn-in is distance that can be added to the threshold braking portion of the corner entry. Even though the radius going into the corner is tighter than normal and requires a slightly lower turn-in speed, this longer threshold braking zone can allow you to wait just a tiny bit longer before going to the brakes.

Going a little bit deeper assures the pass. If the driver on the outside had any lingering thoughts about contesting the corner, watching you dart ahead a half car length as the brakes go on should settle the argument.

Keeping the Position. Now you need to make sure the other driver doesn't get an advantage coming out of the corner. If you early apex the corner and have to be off the power, struggling to get the car to change direction at the corner exit, the cagey competitor will try to give you the corner entry and plan to get on the power earlier than normal, on the perfect line, and get you coming out.

By making the apex, you don't have to worry about hitting your normal throttle application point. The

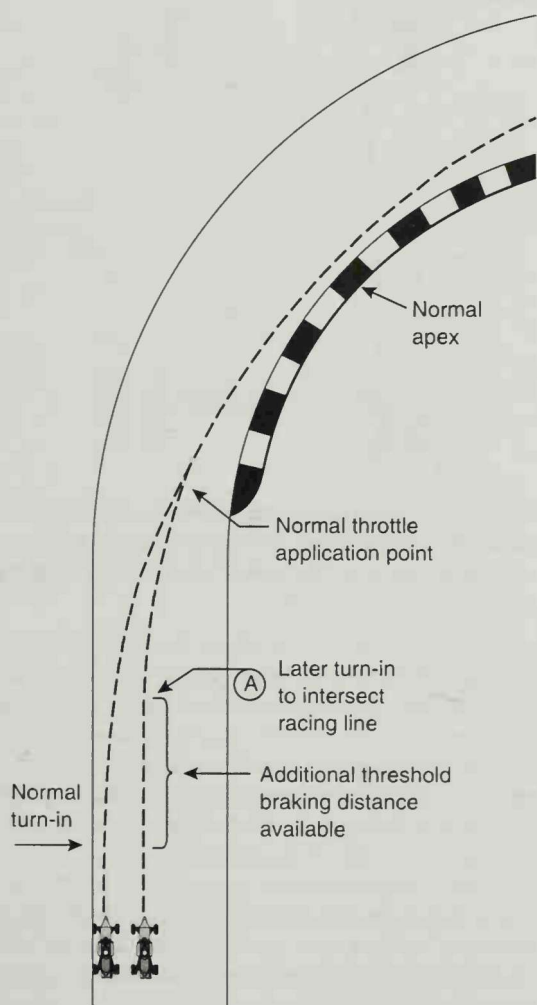


Fig. 9-11. Off-line on the inside, the job of the overtaking car is to get back on line as soon as possible. This will require a later turn-in than a normal lap would call for.

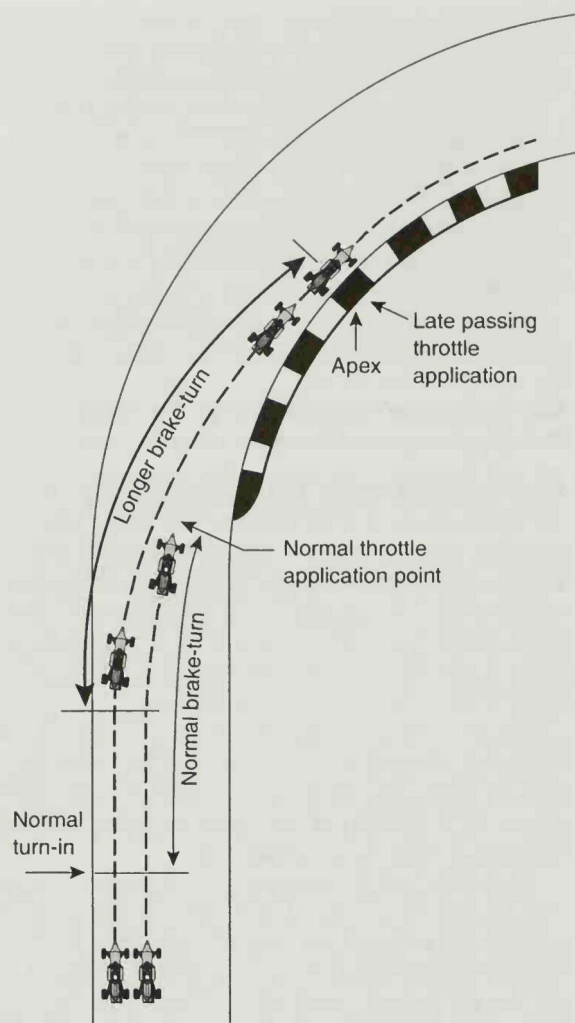


Fig. 9-12. By getting the car back on-line by the apex, the passing car not only changes position, it also determines the throttle application point of the car behind.

driver behind can't accelerate until you do unless he or she wants to drive around the outside of your car. To make the pass and keep it, the key is to determine your opponent's throttle application point which, simply by virtue of human reaction time, is going to be slightly later than yours (see Fig. 9-12).

Passing Briefly: The Double Pass

Look at what happens when you get nervous and turn in early (see Fig. 9-13). Turning at point A, you can run a faster, broad-radius arc down into the corner, easily getting past your competitor on the way in. Once past the apex, though, the path you're on carries you away from the normal racing line, opening up a path for the driver you just passed.

In addition, you'll probably have to still be braking and turning at this point, slowing the car down for the

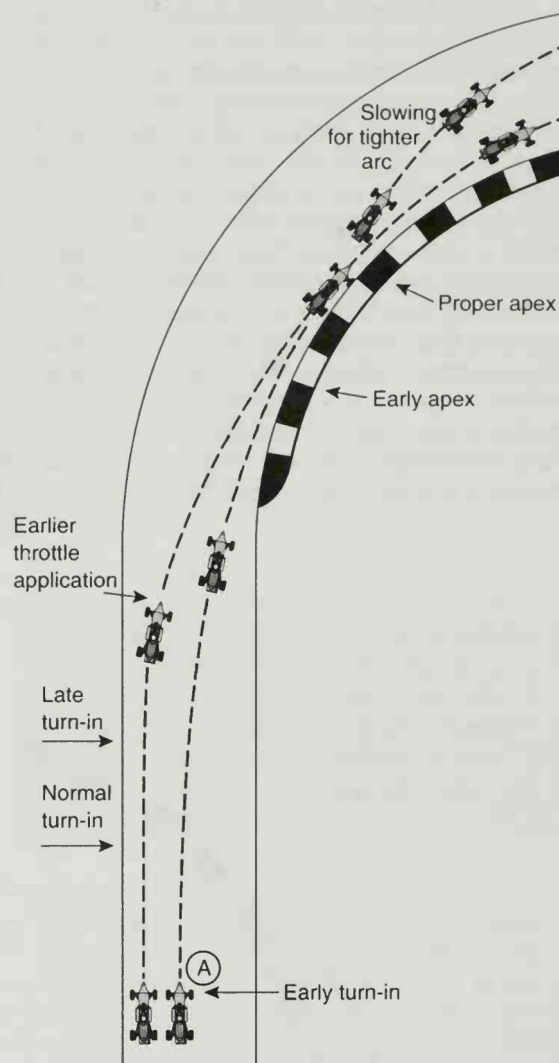


Fig. 9-13. The double pass is very common. It results from the passing car taking an early apex and needing to slow at the exit of the corner to keep the car on the racing surface.

tighter arc you'll have to negotiate late in the corner. Your opponent can see you headed for an early apex and, if anything, turn a little later than normal, concentrating on throttle application. While you're scrambling around on the outside of the road trying to keep your car from going off, he's on the throttle, passing you coming off the corner.

The early apex really kills you going down the next straight. Since the exit arc is tighter and your throttle application point is later, your corner exit speed will be much lower than normal and you'll lose ground all the way to the next corner.

The Early Turn-In Pass

Watch enough races, and you'll see drivers successfully pass by taking a broad arc into the corner inside

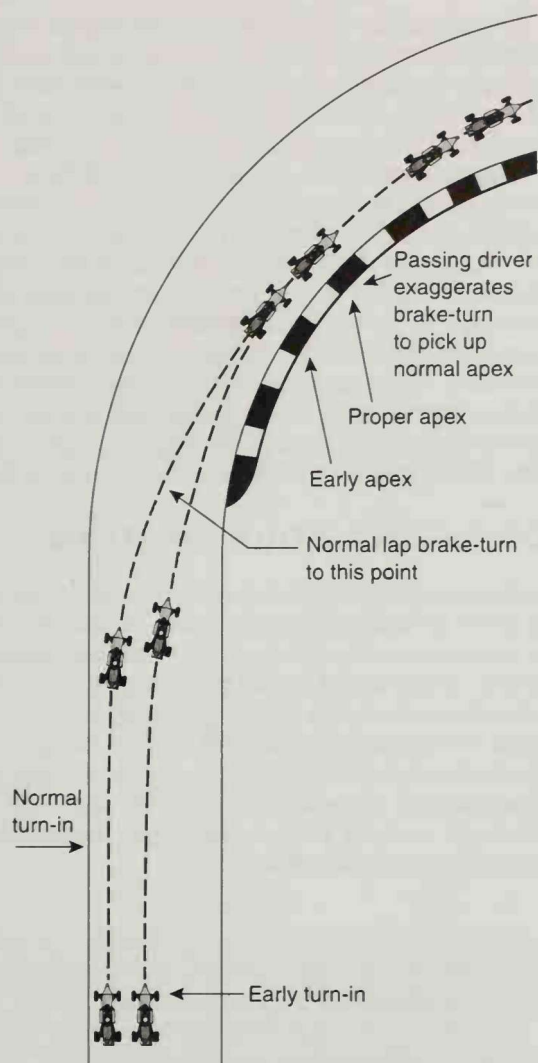


Fig. 9-14. An early turn-in can work if the passing driver is adept at braking and turning and is able to be back on line for the second part of the corner.

their competitor. It looks like an early apex coming, but if the driver can brake and turn all the way to the apex and get back on-line for the exit, it works out (see Fig. 9-14).

The key is being on-line coming out, maintaining your exit speed potential and thereby dictating your competitor's throttle application point. Do this and the pass is made and kept.

THE OUT-BRAKING GAME

If the pass for position comes long before the end of the race, the driver you pass will probably not fight tooth and nail to keep you behind. When it gets toward the end of the race, however, you can't necessarily count on getting cooperation.

"If someone's pressuring you and trying to pass you, you can always brake at a point where you feel it's conservative, then release some brake pressure and coast up alongside him. You're safely, effectively doing the same thing as late braking, while still leaving a margin for error. You're matching his speed at the corner entry."

—Dorsey Schroeder

Waiting Each Other Out

When you're fighting for position, the contest may boil down to who brakes last and who gets to the apex first. When you're side by side on the approach to the corner, the driver on the outside has a tough job defending the position, but it's possible. If the outside driver brakes later than the driver on the inside and, at the turn-in point, is a car-length ahead, the outside driver can go ahead and turn in, successfully turning back the passing attempt. Although the car making the passing attempt established rights to the line by getting alongside in the braking zone, you can't expect the other driver to just roll over. You have to go at least as late on the brakes as the driver you're attempting to pass. You can count on the fact that the other driver will try to brake later than normal if he or she is a tough competitor.

In a way, outbraking passes are like a game of chicken—who's going to lift first. You can't, however, just decide to key off the other driver, not braking until you see the other car start to slow. The other driver may have decided to do the same thing and be waiting for you to brake. The result could be that both of you go hopelessly late: too late to make the corner.

Point of No Return

Either driver, inside or outside, can try to wait for the other to brake, but if you're going to play this game, you should have a point beyond which you won't go.

"Racing somebody down into the braking zone, you think, 'I'm not going to brake 'til he does.' But you do that a couple of times and go sailing down the escape road and lose a ton of time and you realize that you have to draw the line for yourself. I'll go deeper than normal, sometimes a lot deeper, but there's a deepest point I'll go regardless of what the other guy does. Just because he does it doesn't mean you can. The other car can be lighter, have better brakes, better tires. Sometimes you run out of talent before he does."

—David Loring

If the driver on the outside, for example, goes just 10 feet past the point where it becomes impossible to make the corner—the driver on the inside wins. It's an error you're bound to make if you're on the outside and you just blindly try to brake after the driver on the inside does: you leave yourself wide open.

If, on the other hand, you want to hold off the inside pass attempt and through better skill or more experience you go to a last ditch point which is later than that of the driver on the inside, picking up a car length, you'll beat the driver on the inside to the apex. Since the driver on the inside had a "latest point" that was earlier than yours, he or she lost the opportunity to pass but didn't get pressured into going impossibly late and crashing as a result.

Vulnerability

If you're on the outside, you're in more of a position to be a victim of a late-braking mistake. If you go too deep and can't make the corner, at least you don't get in the way of the car on the inside. If you're on the inside and you decide to go deeper, there's a chance that you'll brake too hard, lock up the front tires, and be unable to turn when you both get to the turn-in point. If the outside driver turns in without looking—bang.

If you're the outside driver in this situation, even if you don't turn in, your options are limited to following along on the outside until you *can* turn in behind the other car. Sometimes you can't, and the car on the inside just escorts you down the escape road, if there is one. Without an escape road, you get escorted into the boonies.

How Deep is Deep?

As two Formula Dodges approach a corner like the Sebring Hairpin, they are decelerating from around 110 m.p.h. down to the neighborhood of 40 m.p.h. at the turn-in. Either driver, outside or inside, would like to gain one car length over the other between the brake point and the turn-in. The cars are only 12 feet long, so the battle is to gain 12 feet over the distance of the braking zone.

A Formula Dodge in Race Series trim decelerates at slightly under 1.2 G and, at this rate, it's going to take 2.6 seconds and 290 feet to get slowed from 110 m.p.h. to 40 m.p.h. Since either car needs a 5 foot-per-second advantage to gain 12 feet in 2.6 seconds, you can pull this off by braking .125 seconds (1/8 second) later than the other car. At the approach speed of 110 m.p.h., that means about 20 feet later.

By going 20 feet later on the brakes, the overtaking driver covers the braking zone at an average speed about 3 m.p.h. faster than the other car. This extra 3 m.p.h. can be problematic at the turn-in point. The only way to make the turn-in is to extend the straight line braking zone, knocking off some of the extra speed, and exaggerating the brake-turning effort at the corner entry.

In order to slow the car down to the normal corner entry speed, you would have to come up with an increase in braking ability of almost 5%—not likely if you're already running close to the limit. Going 20 feet deeper is going to take a lot of work and skill.

Standing by the side of the racetrack, 20 feet doesn't look like a lot of distance but, given drivers of equal threshold braking skill, the 20-foot difference in the brake point can mean the difference between winning and finishing second. Few drivers can make up a car-length in the braking zone alone and this is a reflection of the level of parity that you find at the top of each field. When running with the fastest drivers, a 20-foot change in the braking point isn't going to be easy to find, since everyone is bumping up against the edge of the latest possible brake point on normal laps.

DEFENDING YOUR POSITION

In an outbraking situation, you can challenge the other driver to match your brake point. You can test the commitment to the pass by trying to hang alongside on the outside line. Another tactic is not to give the other driver the inside line in the first place. You could drive down the side of the straight that leads to the inside edge of the corner, forcing the overtaking driver to draw alongside on the outside edge of the racetrack, as in Fig. 9-15. At the corner, the outside car ends up being at a disadvantage.

"Recently in the Pro Series...we had one driver who could really go around pretty well all by himself—when he would get into the defensive driving mode, he'd manage to hold somebody behind him, but seven other guys would catch them both. He'd end up much worse off than if he'd just let one guy by. Instead of going second place to third place, he'd go second to ninth."

—Skip Barber

Sometimes, however, this can backfire. When you're on the inside, you have to slow enough to keep the car on the tight radius around the first half of the corner. If you're on the outside, you could brake slightly earlier than normal, drive on the right line for the corner and time it so that you're closing on the car on the inside past the apex, where an open passing lane would open up for the pass on the corner exit.

Surprisingly, this doesn't happen very much. Seeing a car drive defensively up the inside tends to befuddle the driver on the outside and more often than not, they give up the passing attempt. It can be done, but most drivers don't like to take the chance of being hung out to dry on the outside.

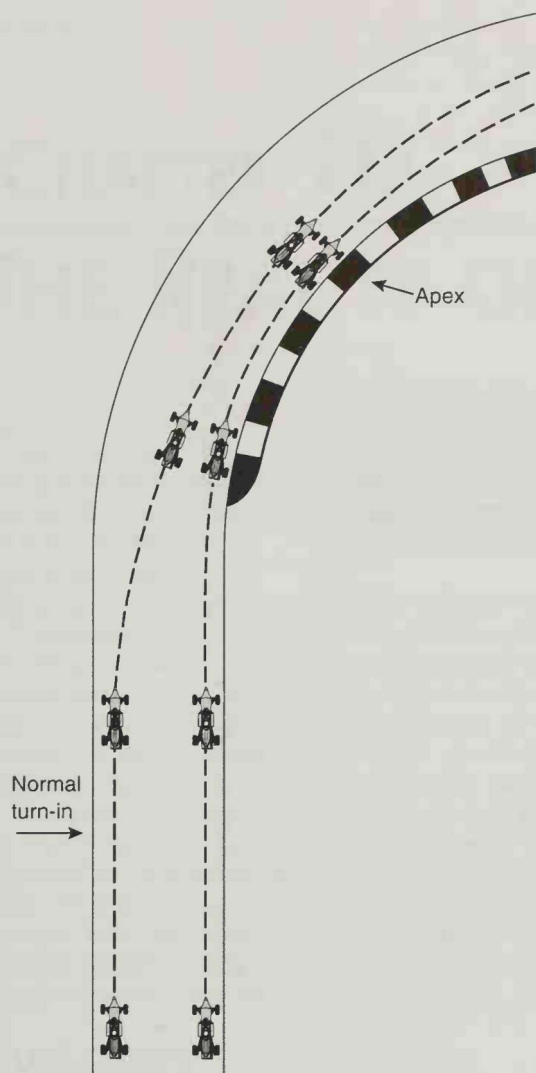


Fig. 9-15. Driving defensively is different than blocking. You can choose to drive up to the corner on the inside edge of the road and challenge the other driver to make an outside pass.

As for the ethics of this technique, it's not really considered blocking, so it's not a dirty move. Any driver has the right to decide on the path they'll take around the racetrack. Driving down the inside of the straight in order to make a pass more difficult for your

competitor—as long as you make the decision to go for the inside *before* the overtaking driver starts to pass on that side—is okay. But if the following car has to get out of the throttle to avoid running into you or has to run off the road because you moved in reaction to a pass attempt, then it's a block.

"There's a time when you want to drive defensively, but I don't believe in this weaving that you see. You can...hurt the momentum of another driver, just occupying a space he wants, and this is especially true on ovals. They have a rule now in Indy Car that you get to move one time. You move a second time and you get black-flagged, and I think that's right. You don't go messing around too much because, especially with open wheelers, they're so easy to get up in the air. People can get killed, and have, and it doesn't take much."

—Danny Sullivan

Blocking

Blocking—weaving down the straight with your eyes on the mirrors, trying to cut off the driver behind—qualifies you as a lower form of life. Like any other sport, racing requires sportsmanship, and if the only way some racers can win is by weaving in front of faster drivers, perhaps they ought to get out of racing and do something more in keeping with their personalities—we hear professional wrestling pays well.

"As professional drivers, I feel that we're all supposed to maintain a certain integrity, a certain honor. I feel like I'm going to race with you just as hard as I possibly can, but if you're driving a better car or you're driving better on that particular day, then you deserve to be in front of me. That doesn't mean I'm going to let you go by easy, but it means if you outdo me, then you're going to get the line. All that I can do is kick myself in the butt and figure out how you did it and get you back."

—Brian Till

